

Chapter 22: Perpendicular Relations in Space (Quan hệ Vuông góc trong Không gian)

Chapter Overview

Perpendicular relations in three-dimensional space are fundamental to understanding spatial geometry. This chapter explores three key perpendicular relationships: **angle between two lines**, **line perpendicular to a plane**, and **two perpendicular planes**.

These concepts are essential for analyzing the structure of buildings, calculating distances and angles in 3D, and solving geometric problems involving pyramids, prisms, and other spatial figures. The **Three Perpendiculars Theorem** provides a powerful tool for proving perpendicularity in space.

Real-world connections:

- Architecture:** Vertical columns perpendicular to horizontal floors
- Engineering:** Perpendicular supports in bridge construction
- Navigation:** Vertical plumb lines and horizontal reference planes
- Manufacturing:** Perpendicular surfaces in machined parts
- Astronomy:** Earth's axis and orbital plane

22.1 Bilingual Vocabulary (Từ vựng Song ngữ)

Core Concepts

English	IPA	Vietnamese	Example
Perpendicular	/ˌpɜːrpənˈdɪkjələr/	Vuông góc	<i>Two lines are perpendicular if they meet at a right angle.</i>
Right angle	/raɪt ˈæŋɡəl/	Góc vuông	<i>A right angle measures 90 degrees.</i>

Angle between lines	<i>/ˈæŋɡəl bɪˈtwiːn laɪnz/</i>	Góc giữa hai đường thẳng	<i>The angle between two skew lines is defined using parallel translations.</i>
Projection	<i>/prəˈdʒekʃən/</i>	Hình chiếu	<i>The projection of a point onto a plane is found by dropping a perpendicular.</i>

Line and Plane Perpendicularity

English	IPA	Vietnamese	Example
Line perpendicular to plane	<i>/laɪn ˌpɜːrpənˈdɪkjələr tə pleɪn/</i>	Đường thẳng vuông góc với mặt phẳng	<i>A line is perpendicular to a plane if it is perpendicular to every line in the plane.</i>
Orthogonal projection	<i>/ɔːrˈθɒɡənəl prəˈdʒekʃən/</i>	Phép chiếu vuông góc	<i>Orthogonal projection maps points perpendicularly onto a plane.</i>
Foot of perpendicular	<i>/fʊt əv ˌpɜːrpənˈdɪkjələr/</i>	Chân đường vuông góc	<i>The foot of the perpendicular is where the line meets the plane.</i>

Plane Perpendicularity

English	IPA	Vietnamese	Example
Perpendicular planes	<i>/ˌpɜːrpənˈdɪkjələr pleɪnz/</i>	Hai mặt phẳng vuông góc	<i>Two planes are perpendicular if the angle between them is 90°.</i>

Dihedral angle	/daɪˈhiːdrəl ˈæŋɡəl/	Góc nhị diện	<i>A dihedral angle is the angle between two planes.</i>
Plane angle of dihedral	/pleɪn ˈæŋɡəl əv daɪˈhiːdrəl/	Góc phẳng của góc nhị diện	<i>The plane angle is measured in a plane perpendicular to the edge.</i>

Angles and Distances

English	IPA	Vietnamese	Example
Angle between line and plane	/ˈæŋɡəl bɪˈtwiːn laɪn ənd pleɪn/	Góc giữa đường thẳng và mặt phẳng	<i>The angle is measured between the line and its projection.</i>
Distance from point to plane	/ˈdɪstəns frəm pɔɪnt tə pleɪn/	Khoảng cách từ điểm đến mặt phẳng	<i>The distance is the length of the perpendicular segment.</i>
Three Perpendiculars Theorem	/θriː ˌpɜːrpən ˈdɪkjələrz ˈθiərəm/	Định lý ba đường vuông góc	<i>This theorem relates perpendicularity in space to perpendicularity in a plane.</i>

22.2 Angle Between Two Lines in Space

Definition

Angle Between Two Lines

The angle between two lines

a

and

b

in space is defined as follows:

1. Choose any point

O

in space

2. Draw lines

a'

and

b'

through

O

parallel to

a

and

b

respectively

3. The angle between

a

and

b

is the angle between

a'

and

b'

Note: This angle is well-defined (independent of the choice of

O

) and satisfies

$$0^\circ \leq \theta \leq 90^\circ$$

.

Convention: We always take the acute angle (or right angle) between the lines.

Perpendicular Lines

Perpendicular Lines in Space

Two lines

a

and

b

are **perpendicular** (written

$a \perp b$

) if the angle between them is

90°

.

Important: Perpendicular lines in space do not need to intersect—they can be skew lines.

22.3 Line Perpendicular to a Plane

Definition

Line Perpendicular to Plane

A line

d

is **perpendicular to a plane**

(P)

(written

$d \perp (P)$

) if

d

is perpendicular to **every line** in plane

(P)

.

This is a very strong condition—perpendicular to ALL lines in the plane, not just some lines.

Criterion for Line Perpendicular to Plane

Sufficient Condition

A line

d

is perpendicular to plane

(P)

if and only if

d

is perpendicular to **two intersecting lines** in

(P)

.

In symbols: If

$a, b \subset (P)$

,

$a \cap b \neq \emptyset$

, and

$d \perp a$

,

$d \perp b$

, then

$d \perp (P)$

.

Proof strategy: To show

$d \perp (P)$

, find two intersecting lines in

(P)

and show

d

is perpendicular to both.

Properties of Line Perpendicular to Plane

Property 1: Uniqueness

Through a given point, there is **exactly one line** perpendicular to a given plane.

Property 2: Uniqueness of Perpendicular from External Point

From a point

A

not on plane

(P)

, there is **exactly one point**

H

on

(P)

such that

$AH \perp (P)$

.

Point

H

is called the **orthogonal projection** (or **foot of perpendicular**) of

A

onto

(P)

.

Property 3: Parallel Lines

- If line
 $a \perp (P)$
and
 $b \parallel a$
, then
 $b \perp (P)$

- If
 $a \perp (P)$
and
 $b \perp (P)$
, then
 $a \parallel b$

In words: Lines perpendicular to the same plane are parallel to each other.

22.4 Orthogonal Projection

Definition

Orthogonal Projection

The **orthogonal projection** of a point

M

onto plane

(P)

is the foot

H

of the perpendicular from

M

to

(P)

.

The **orthogonal projection** of a figure

F

onto

(P)

is the set of orthogonal projections of all points of

F

.

Properties of orthogonal projection:

- Projection of a point is a point
 - Projection of a line (not perpendicular to the plane) is a line
 - Projection of a line perpendicular to the plane is a point
 - Projection preserves collinearity and betweenness
-

22.5 Three Perpendiculars Theorem

The Theorem

Three Perpendiculars Theorem

Let

d

be a line not perpendicular to plane

(P)

, and let

d'

be the orthogonal projection of

d

onto

(P)

.

A line

b

in plane

(P)

is perpendicular to

d

if and only if

b

is perpendicular to

d'

.

In symbols:

$$b \perp d \Leftrightarrow b \perp d'$$

(where

$b \subset (P)$

)

Significance: This theorem allows us to check perpendicularity between a line and its projection in the plane, which is often easier than checking perpendicularity in 3D.

Typical application: To prove

$$b \perp d$$

(where

$$d$$

is not in

$$(P)$$

), we:

1. Find the projection

$$d'$$

of

$$d$$

onto

$$(P)$$

2. Prove

$$b \perp d'$$

(in the plane)

3. Conclude

$$b \perp d$$

22.6 Angle Between Line and Plane

Definition

Angle Between Line and Plane

Let

$$d$$

be a line and

$$(P)$$

a plane. The **angle between**

$$d$$

and

$$(P)$$

is defined as:

- If $d \perp (P)$, the angle is 90°
- If $dot \perp (P)$, let d' be the projection of d onto (P) . The angle between d and (P) is the angle between d and d'

Range:

$$0^\circ \leq \theta \leq 90^\circ$$

Note: The angle between a line and a plane is the **smallest** angle between the line and any line in the plane.

Computing the Angle

To find the angle

θ

between line

d

and plane

(P)

:

1. Find point

A

on

d

2. Find the orthogonal projection

H

of

A

onto

(P)

3. Find another point

B

on

d

and its projection

B'

onto

(P)

4. The angle

$\angle AHB'$

(or

$\angle BAH$

) is the angle between

d

and

(P)

Formula: If

AH

is the perpendicular distance and

AB

is a segment on the line:

$$\sin \theta = \frac{AH}{AB}$$

22.7 Perpendicular Planes

Angle Between Two Planes

Angle Between Two Planes

The **angle between two planes**

(P)

and

(Q)

is defined as the angle between two lines

a

and

b

where:

- $a \perp (P)$

- $b \perp (Q)$

This angle is well-defined (independent of the choice of

a

and

b

) and satisfies

$$0^\circ \leq \theta \leq 90^\circ$$

.

Perpendicular Planes

Perpendicular Planes

Two planes

(P)

and

(Q)

are **perpendicular** (written

$$(P) \perp (Q)$$

) if the angle between them is
 90°

.

Criterion for Perpendicular Planes

Sufficient Condition

Two planes

(P)

and

(Q)

are perpendicular if and only if one plane contains a line perpendicular to the other plane.

In symbols:

$$(P) \perp (Q) \Leftrightarrow$$

there exists

$$d \subset (P)$$

such that

$$d \perp (Q)$$

Dihedral Angle

Dihedral Angle

A **dihedral angle** is the figure formed by two half-planes (called **faces**) sharing a common edge.

The **plane angle** of a dihedral angle is measured in any plane perpendicular to the edge, as the angle between the two lines where this plane intersects the two faces.

Note: All plane angles of the same dihedral angle are equal.

22.8 Worked Examples

Example 1: Proving Line Perpendicular to Plane

Problem: Given pyramid

$S.ABC$

with

$SA \perp AB$

,

$SA \perp AC$

, and

$AB \perp AC$

. Prove that

$SA \perp (ABC)$

.

Solution:

We have:

- $SA \perp AB$
- $SA \perp AC$
- AB
and
 AC
are two intersecting lines in plane
 (ABC)
(they intersect at
 A
)

By the criterion for line perpendicular to plane:

$SA \perp (ABC)$

Answer:

$SA \perp (ABC)$

by the two-line criterion.

Example 2: Finding Orthogonal Projection

Problem: Given pyramid

$S.ABCD$

with square base

$ABCD$

and

$SA \perp (ABCD)$

. Find:

a) The projection of

S

onto

$(ABCD)$

b) The projection of edge

SB

onto

$(ABCD)$

c) The projection of triangle

SBC

onto

$(ABCD)$

Solution:

a) Since

$SA \perp (ABCD)$

and

$A \in (ABCD)$

, point

A

is the projection of

S

onto

$(ABCD)$

.

b) The projection of

SB

is the line segment

AB

(connecting the projections of

S

and

B

).

c) The projection of triangle

SBC

is triangle

ABC

(the projections of vertices

S

,

B

,

C

are

A

,

B

,

C

).

Answer: a) Point

A

; b) Segment

AB

; c) Triangle

ABC

Example 3: Using Three Perpendiculars Theorem

Problem: Given pyramid

$S.ABC$

with

$SA \perp (ABC)$
and triangle
 ABC
right-angled at
 B

. Prove that
 $BC \perp SB$

.
Solution:

Since
 $SA \perp (ABC)$
, we have
 $SA \perp BC$
(as
 $BC \subset (ABC)$
).

Also given:
 $AB \perp BC$
(since
 $\angle ABC = 90^\circ$
).

The projection of
 SB
onto plane
 (ABC)
is
 AB

.
By the Three Perpendiculars Theorem:

$BC \perp AB$
and
 $BC \subset (ABC)$
implies
 $BC \perp SB$

.

Answer:

$$BC \perp SB$$

by the Three Perpendiculars Theorem.

Example 4: Finding Angle Between Line and Plane

Problem: Given pyramid

$$S.ABCD$$

with square base

$$ABCD$$

of side

$$a$$

, and

$$SA \perp (ABCD)$$

with

$$SA = a$$

. Find the angle between

$$SC$$

and plane

$$(ABCD)$$

.

Solution:

The projection of

$$SC$$

onto

$$(ABCD)$$

is

$$AC$$

(since

$$A$$

is the projection of

$$S$$

).

The angle between

$$SC$$

and

(ABCD)

is

$\angle SCA$

.

In right triangle

SAC

:

- $SA = a$
(perpendicular to plane)
- $AC = a\sqrt{2}$
(diagonal of square with side a)

$$\tan(\angle SCA) = \frac{SA}{AC} = \frac{a}{a\sqrt{2}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

Therefore:

$$\angle SCA = \arctan\left(\frac{\sqrt{2}}{2}\right) \approx 35.26^\circ$$

Alternatively, using sine:

$$\sin(\angle SCA) = \frac{SA}{SC}$$

$$SC = \sqrt{SA^2 + AC^2} = \sqrt{a^2 + 2a^2} = a\sqrt{3}$$

$$\sin(\angle SCA) = \frac{a}{a\sqrt{3}} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$$

$$\angle SCA = \arcsin\left(\frac{\sqrt{3}}{3}\right) \approx 33.56^\circ$$

Wait, let me recalculate. The angle between

SC

and the plane is

$\angle SCA$

, where

C

is on the plane and

A
is the projection of
 S

Actually, the angle should be

$\angle SCA$
measured at
 C

, not at
 A

Let me reconsider: The angle between line

SC
and plane
 $(ABCD)$

is the angle between

SC
and its projection
 AC

, which is

$\angle SCA$

or

$\angle SAC$

Actually, it's

$\angle SCA$

(at point

C

) or equivalently

$\angle SAC$

(at point

A

). Let's use

$\angle SAC$

:

$$\tan(\angle SAC) = \frac{SA}{AC} = \frac{a}{a\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\angle SAC = \arctan\left(\frac{1}{\sqrt{2}}\right) \approx 35.26^\circ$$

Answer: The angle is
 $\arctan\left(\frac{\sqrt{2}}{2}\right) \approx 35.26^\circ$

Example 5: Distance from Point to Plane

Problem: Given pyramid

$S.ABC$

with

$SA \perp (ABC)$

,

$SA = 6$

cm, and triangle

ABC

has

$AB = 3$

cm,

$AC = 4$

cm,

$BC = 5$

cm. Find the distance from

S

to plane

(ABC)

.

Solution:

Since

$SA \perp (ABC)$

, the distance from

S

to plane

(ABC)

is simply the length

SA

.

Distance

$$= SA = 6$$

cm.

Answer: 6 cm

Example 6: Proving Planes are Perpendicular

Problem: Given cube

$ABCD.A'B'C'D'$

. Prove that plane

$(ACC'A')$

is perpendicular to plane

$(ABCD)$

.

Solution:

We need to show that plane

$(ACC'A')$

contains a line perpendicular to plane

$(ABCD)$

.

Since

$ABCD.A'B'C'D'$

is a cube, we have

$$AA' \perp (ABCD)$$

(vertical edge perpendicular to base).

Also,

$$AA' \subset (ACC'A')$$

(

A

and

A'

are vertices of the plane).

Therefore, by the criterion for perpendicular planes:

$$(ACC'A') \perp (ABCD)$$

Answer:

$$(ACC'A') \perp (ABCD)$$

because

$$AA' \subset (ACC'A')$$

and

$$AA' \perp (ABCD)$$

.

Example 7: Finding Dihedral Angle

Problem: Given pyramid

$$S.ABC$$

with

$$SA \perp (ABC)$$

,

$$SA = 6$$

, and equilateral triangle

$$ABC$$

with side 4. Find the dihedral angle along edge

$$BC$$

(angle between planes

$$(SBC)$$

and

$$(ABC)$$

).

Solution:

To find the dihedral angle along

$$BC$$

, we need to find a plane perpendicular to

$$BC$$

.

Let

$$M$$

be the midpoint of

BC

. In equilateral triangle

ABC

, we have

$$AM \perp BC$$

.

Since

$$SA \perp (ABC)$$

, we have

$$SA \perp BC$$

.

In plane

(SAM)

:

- $AM \perp BC$
(in the base)

- SM
is in the plane

The dihedral angle is

$\angle SMA$

(angle between

SM

and

AM

, both perpendicular to edge

BC

).

Calculate

AM

: In equilateral triangle with side 4:

$$AM = \frac{4\sqrt{3}}{2} = 2\sqrt{3}$$

In right triangle

SAM

:

$$\tan(\angle SMA) = \frac{SA}{AM} = \frac{6}{2\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$$

Therefore:

$$\angle SMA = \arctan(\sqrt{3}) = 60^\circ$$

Answer: The dihedral angle is 60° .

Example 8: Perpendicular Planes Application

Problem: Two perpendicular planes

(P)

and

(Q)

intersect along line

d

. A point

A

is 3 cm from

d

in plane

(P)

, and a point

B

is 4 cm from

d

in plane

(Q)

. Find

AB

.

Solution:

Let

H

be the foot of perpendicular from

A
 to
 d
 , and
 K
 be the foot of perpendicular from
 B
 to
 d
 .

Since
 $(P) \perp (Q)$
 and both planes contain
 d
 :

- $AH \perp d$
 (in plane
 (P)
)
- $BK \perp d$
 (in plane
 (Q)
)
- $AH \perp (Q)$
 (since
 $(P) \perp (Q)$
)

If
 $H = K$
 (same point on
 d
):

$$AB = \sqrt{AH^2 + HB^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$$

cm

Answer:

$$AB = 5$$

cm (assuming

A

and

B

project to the same point on

d

).

22.9 Applications

Application 1: Vertical Construction

Problem: A vertical column is perpendicular to a horizontal floor. If the column is 3 meters tall and a point on the floor is 4 meters from the base of the column, what is the distance from the top of the column to that point?

Solution:

Let

S

be the top of the column,

A

be the base, and

M

be the point on the floor.

Since the column is vertical:

$SA \perp$

(floor)

Given:

$SA = 3$

m,

$AM = 4$

m

In right triangle

SAM

:

$$SM = \sqrt{SA^2 + AM^2} = \sqrt{9 + 16} = 5$$

m

Answer: 5 meters

Application 2: Solar Panel Angle

Problem: A solar panel is tilted at

30°

from the horizontal ground. If the panel is 2 meters long, what is the vertical height of the top edge above the bottom edge?

Solution:

The angle between the panel and the ground is

30°

.

The vertical height

h

is:

$$h = 2 \sin(30^\circ) = 2 \times 0.5 = 1$$

meter

Answer: 1 meter

Application 3: Roof Pitch

Problem: A gabled roof has two rectangular faces meeting at a ridge. Each face makes a

35° angle with the horizontal attic floor. What is the dihedral angle between the two roof faces?

Solution:

Each roof face makes

35°

with the horizontal.

The dihedral angle between the two faces is:

$$180^\circ - 2 \times 35^\circ = 180^\circ - 70^\circ = 110^\circ$$

Answer:

110°

22.10 Practice Exercises

Exercise 1: Given pyramid

$S.ABCD$

with rectangular base and

$SA \perp (ABCD)$

. Prove that:

a)

$AB \perp (SAD)$

b)

$BC \perp (SAB)$

Exercise 2: Given cube

$ABCD.A'B'C'D'$

with edge length

a

. Find the angle between diagonal

AC'

and plane

$(ABCD)$

.

Exercise 3: Given tetrahedron

$OABC$

with

OA

,

OB

,

OC

mutually perpendicular, each of length

a

. Find the distance from

O

to plane

(ABC)

Exercise 4: Prove that if two planes are both perpendicular to the same line, then they are parallel.

Exercise 5: Given pyramid

$S.ABC$

with

$$SA = SB = SC$$

and

$$SA \perp SB$$

,

$$SB \perp SC$$

,

$$SC \perp SA$$

. Prove that the projection of

S

onto

(ABC)

is the circumcenter of triangle

ABC

.

22.11 Chapter Summary

Key Concepts

1. Angle Between Two Lines

- Defined using parallel translations
- Range:
 $0^\circ \leq \theta \leq 90^\circ$
- Perpendicular lines:
 $\theta = 90^\circ$

2. Line Perpendicular to Plane

- **Criterion:** Perpendicular to two intersecting lines in the plane

- **Properties:**

- Unique perpendicular through a point
- Parallel lines perpendicular to same plane
- Lines perpendicular to same plane are parallel

3. Three Perpendiculars Theorem

- $b \perp d \Leftrightarrow b \perp d'$
(where d' is projection of d)
- Powerful tool for proving perpendicularity

4. Angle Between Line and Plane

- Angle between line and its projection
- Range:
 $0^\circ \leq \theta \leq 90^\circ$
- Formula:
 $\sin \theta = \frac{h}{l}$
(height over length)

5. Perpendicular Planes

- **Criterion:** One plane contains line perpendicular to the other
- **Dihedral angle:** Measured in plane perpendicular to edge
- **Properties:** Transitivity, symmetry

22.12 Review Exercises

Exercise 1: Given pyramid

$S.ABCD$

with square base

$ABCD$

of side

a

and

$$SA \perp (ABCD)$$

with

$$SA = a\sqrt{2}$$

a) Find the angle between

$$SC$$

and

$$(ABCD)$$

b) Find the angle between

$$SB$$

and

$$(SAC)$$

c) Prove that

$$(SBD) \perp (SAC)$$

Exercise 2: Given rectangular box

$$ABCD.A'B'C'D'$$

with

$$AB = 3$$

,

$$BC = 4$$

,

$$AA' = 5$$

.

a) Prove that

$$AC \perp (BDD'B')$$

b) Find the angle between

$$A'C$$

and

$$(ABCD)$$

c) Find the distance from

$$A$$

to plane

$$(A'BD)$$

Exercise 3: Given tetrahedron

$ABCD$

with

$$AB = AC = AD = a$$

and

$$\angle BAC = \angle CAD = \angle DAB = 90^\circ$$

.

a) Prove that

$$AB \perp (ACD)$$

b) Find the distance from

A

to plane

$$(BCD)$$

c) Find the angle between

$$(ABC)$$

and

$$(BCD)$$

Exercise 4: Prove that the locus of points equidistant from two given points

A

and

B

is the plane perpendicular to

AB

at its midpoint.

Exercise 5: Given two perpendicular planes intersecting along line

d

. A line

a

in one plane makes a

$$30^\circ$$

angle with

d

. Find the angle between

a

and the other plane.

22.13 Memory Aids

Line Perpendicular to Plane

"Two friends make it perpendicular"

To show

$$d \perp (P)$$

: Find **two intersecting** lines in

(P)

and show

d

is perpendicular to both.

Three Perpendiculars Theorem

"Check the shadow"

To prove

$$b \perp d$$

: Check if

b

is perpendicular to the **projection** (shadow) of

d

on the plane.

Angle Between Line and Plane

"Line to shadow angle"

The angle between a line and a plane is the angle between the **line and its projection** on the plane.

Perpendicular Planes

"One line does the job"

To show

$$(P) \perp (Q)$$

: Find **one line** in

(P)

that is perpendicular to

(Q)

22.14 Common Mistakes to Avoid

⚠ **Mistake 1:** Thinking perpendicular lines must intersect

- **Wrong:** If

$$a \perp b$$

, then

a

and

b

intersect

- **Correct:** Perpendicular lines in space can be skew

⚠ **Mistake 2:** Using only one line to prove line perpendicular to plane

- **Wrong:**

$$d \perp a$$

and

$$a \subset (P)$$

, therefore

$$d \perp (P)$$

- **Correct:** Need

d

perpendicular to **two intersecting lines** in

(P)

⚠ **Mistake 3:** Confusing angle between line and plane with angle between two lines

- The angle between a line and a plane is measured to the **projection**, not to an arbitrary line in the plane

⚠ **Mistake 4:** Forgetting the Three Perpendiculars Theorem applies only to projections

- The theorem relates perpendicularity to the **projection** of the line, not to arbitrary lines

⚠ **Mistake 5:** Assuming perpendicular planes intersect at right angles everywhere

- Perpendicular planes intersect along a line; the 90° angle is measured in a plane **perpendicular to the edge**
-

🎉🎉 **CONGRATULATIONS!** 🎉🎉

You have completed **Chapter 22: Perpendicular Relations in Space** — the **FINAL CHAPTER** of this comprehensive mathematics textbook!

You now have a complete understanding of **spatial geometry**, including:

- Parallel and perpendicular relationships in 3D
- Angles and distances in space
- Coordinate systems and equations
- Conic sections and spatial figures

This marks the completion of 22 chapters covering Algebra, Calculus, and Geometry! 🚀📖🌟